

Ryotaro Okabe

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EDUCATION

Ph.D. in Chemistry, Massachusetts Institute of Technology, *MA, USA* Sept. 2021–Present, GPA: 4.3/5.0

M.S. in Engineering, Tokyo Institute of Technology, *Japan* / Sept. 2019–Sept. 2021, GPA: 3.63/4.50

B.S. in Engineering, Tokyo Institute of Technology, *Japan* / Apr. 2016–Sept. 2019, GPA: 3.92/4.00 (*Early Graduation*)

- **Awards:** Graduate Scholarships (IBM PhD Fellowship Awards, Heiwa Nakajima Foundation, Nakatani Foundation, and JASSO), MRS2025S Graduate Student Silver Award, MRS2025S Best Poster Award, Nakatani Foundation's Fully-funded research internship (Yale University, Rice University), NERSC GenAI AY2024 Resource Allocations, CHRNS Summer School on Neutron Scattering (NIST), NMR Society of Japan Young Scientists Poster Award
- **Courses:** Quantum mechanics, Statistical mechanics, Solid state physics, Machine learning, Group theory
- **Teaching Assistant:** Quantum Mechanics (2021), General Chemistry (2022)
- **Leadership:** Japanese Association of MIT Treasurer

SKILLS

- **Programming:** Python, CUDA, MATLAB, Git, Scripting (Bash), LaTeX, HTML, Vim, Tcl
- **Software:** Machine learning (PyTorch, TensorFlow, Scikit-learn, PyTorch Lightning, WandB, Transformers), Material simulations (ABINIT, Quantum Espresso, Wannier90, Phonopy, Phono3py, OpenMC, Kwant), NMR simulations (SIMPSON, SpinEvolution), Linux, Origin, ChemDraw
- **Material characterization:** Solid-state NMR, Powder X-ray diffraction, Metal-organic framework synthesis

EXPERIENCE

Research Assistant, Massachusetts Institute of Technology (Advisor: Prof. Mingda Li), *MA, USA* / Nov. 2021–Present

- Developed machine learning methods to explore the structure-property relationship in materials (e.g., prediction of phonon dispersion from material structures, generating quantum material candidates).
- Developed machine learning methods to localize the radiation sources with detector arrays.

Research Intern, IBM Thomas J. Watson Research Center (Mentor: Dr. Ching-Tzu Chen), *NY, USA* / May– Aug. 2024

- High-throughput calculation of electrical conductance in topological interconnect materials.
- Developed machine learning methods to predict the conductance of slab materials.

Research Assistant, Tokyo Institute of Technology (Advisor: Prof. Yoshitaka Ishii), *Japan* / Oct. 2018–Aug. 2021

- Implemented new schemes of solid-state nuclear magnetic resonance (SSNMR) for selective signal acquisition and evaluated them through multidimensional SSNMR experiments with numerical simulations.

Research Intern, Yale University (Advisor: Prof. Charles A. Schmuttenmaer), *CT, USA* / Jan.–Apr. 2019

- Synthesized conditions of metal-organic frameworks (MOFs) as photosensitizers and characterized them using powder X-ray diffraction, electrochemical experiments, and time-resolved terahertz spectroscopy.

Research Intern, Rice University (Advisor: Prof. Angel A. Martí), *TX, USA* / Aug.–Sept. 2018

- Observed the aggregation kinetics of Amyloid beta (A β) protein and interactions with the metal complex.

PUBLICATIONS

1. B Han†, **R Okabe**†, A Chotrattanapituk†, M Cheng†, M Li and Y Cheng, “AI-Powered Exploration of Molecular Vibrations, Phonons and Spectroscopy.” [Digital Discovery](#), **2025**.
2. M Cheng*†, C Fu†, **R Okabe**†, A Chotrattanapituk†, A Boonkird, NT Hung and M Li, “AI-driven materials design: a mini-review,” [arXiv:2502.02905](#), **2025**.
3. **R Okabe**†, Z West†, A Chotrattanapituk, M Cheng, D Cordova Carrizales, W Xie, RJ Cava and M Li, “Large Language Model-Guided Prediction Toward Quantum Materials Synthesis,” [arXiv:2410.20976](#), **2024**.
4. **R Okabe**, M Cheng, A Chotrattanapituk, NT Hung, X Fu, B Han, Y Wang, W Xie, RJ Cava, TS Jaakkola, Y Cheng, M Li, “Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates,” [arXiv:2407.04557](#), **2024**.

5. P Siriviboon[†], C Fu[†], M Landry[†], **R Okabe**, D Cordova Carrizales, Y Wang and M Li, “Quantum Theory of X-ray Photon Correlation Spectroscopy,” [arXiv:2412.03635](https://arxiv.org/abs/2412.03635), **2024**.
6. NT Hung, **R Okabe**, A Chottrattanapituk and M Li, “Ensemble-Embedding Graph Neural Network for Direct Prediction of Optical Spectra from Crystal Structure,” [Adv. Mater., 2409175](https://doi.org/10.1002/adma.2409175), **2024**.
 - Press release at [Tohoku University Research News](https://tohoku.ac.jp/en/research-news/graphical-abstract), [Graphical Abstract](https://doi.org/10.1002/adma.2409175).
7. **R Okabe**[†], A Chottrattanapituk[†], A Boonkird, N Andrejevic, X Fu, TS Jaakkola, Q Song, T Nguyen, NC Drucker, S Mu, B Liao, Y Cheng, M Li, “Virtual Node Graph Neural Network for Full Phonon Prediction,” [Nat Comput Sci 4, 522–531](https://doi.org/10.1002/nat.2024.522-531), **2024**.
 - Journal [Research Briefing](https://doi.org/10.1002/nat.2024.522-531), July 2024 issue [Front Cover](https://doi.org/10.1002/nat.2024.522-531), press release at [MIT News](https://news.mit.edu/2024/quantum-material-discovery-07).
8. M Mandal, A Chottrattanapituk, K Woller, H Xu, N Mao, **R Okabe**, A Boonkird, T Nguyen, NC Drucker, T Momiki, J Li, J Kong, and M Li, “Precise Fermi-level engineering in a topological Weyl semimetal via fast ion implantation,” [Appl. Phys. Rev. 11, 021429](https://doi.org/10.1063/1.51429), **2024**.
 - Press release at [MIT News](https://news.mit.edu/2024/quantum-material-discovery-07).
9. M Cheng, **R Okabe**, A Chottrattanapituk, M Li, “Machine Learning Detection of Majorana Zero Modes from Zero Bias Peak Measurements,” [Matter 7, 2507](https://doi.org/10.1038/s41535-024-02507-2), **2024**.
10. **R Okabe**[†], S Xue[†], J Vavrek[†], J Yu, R Pavlovsky, V Negut, B Quiter, J Cates, T Liu, B Forget, S Jegelka, G Kohse, L-W Hu, M Li, “Tetris-Inspired Detector with Neural Network for Radiation Mapping,” [Nat. Commun. 15, 3061](https://doi.org/10.1038/s41535-024-03061-2), **2024**.
 - Press release at [MIT News](https://news.mit.edu/2024/quantum-material-discovery-07), [Physics World](https://www.physicsworld.com/a/quantum-material-discovery-07), [Nuclear Engineering International](https://www.nuclear-engineering-international.com/news/quantum-material-discovery-07).
11. NC Drucker, T Nguyen, M Mandal, P Siriviboon, Y Quan, A Boonkird, **R Okabe**, F Li, K Burrage, F Funuma, M Matsuda, D Abernathy, T Williams, S Chi, F Ye, C Nelson, B Liao, P Volkov, and M Li, “Incipient Nematicity from Electron Flat Bands in a Kagome Metal”, [arXiv:2401.17141](https://arxiv.org/abs/2401.17141), **2024**
12. **R Okabe**, M Li, Y Iwasaki, N Regnault, C Felser, M Shirai, A Kovacs, T Schrefl, A Hirohata. "Materials Informatics for the Development and Discovery of Future Magnetic Materials", [IEEE Magnetic Letters, 14, 1-5](https://doi.org/10.1109/MAG.2023.1000001), **2023**.
 - IEEE Magnetics Letters (Volume: 14) [Front Cover](https://doi.org/10.1109/MAG.2023.1000001).
13. M Mandal, NC Drucker, P Siriviboon, T Nguyen, A Boonkird, TN Lamichhane, **R Okabe**, A Chottrattanapituk and M Li, “Topological superconductors from a materials perspective”, [Chem. Mater. 35, 16, 6184–6200](https://doi.org/10.1002/chem.202300161), **2023**.
14. NC Drucker, T Liu, Z Chen, **R Okabe**, A Chottrattanapituk, T Nguyen, Y Wang, M Li, “Challenges and Opportunities of Machine Learning on Neutron and X-ray Scattering”, [Synchrotron Radiat. News, 35, 16-20](https://doi.org/10.1002/syn.202200161), **2022**.
15. HA Merker, H Heiberger, L Nguyen, T Liu, Z Chen, N Andrejevic, NC Drucker, **R Okabe**, Y Wang, T Smidt, M Li, “Machine Learning Magnetism Classifiers from Atomic Coordinates” [iScience, 25, 10, 105192](https://doi.org/10.1002/isce.202200161), **2022**.
16. T Matsunaga, **R Okabe**, Y Ishii, “Efficient solvent suppression with adiabatic inversion for ¹H-detected solid-state NMR” [J. Biomol. NMR. 12, 365-370](https://doi.org/10.1002/jbm.202100161), **2024, 2021**.
17. B Pattengale, JG Freeze, M Guberman-Pfeffer, **R Okabe**, S Ostresh, S Chaudhuri, VS Batista, CA Schmittenmaer, “A Conductive Metal-Organic Framework Photoanode,” [Chem. Sci. 11, 9593–9603](https://doi.org/10.1002/chem.202000161), **2020**.
 - [Conference abstract proceedings \(2022\)](https://doi.org/10.1002/chem.202000161).
18. B Pattengale, J Neu, S Ostresh, G Hu, JA. Spies, **R Okabe**, GW Brudvig, CA Schmittenmaer. “Metal-Organic Framework Photoconductivity via Time-Resolved Terahertz Spectroscopy,” [J. Am. Chem. Soc. 141, 9793–9797](https://doi.org/10.1021/ja00161a001), **2019**.

PRESENTATIONS

1. **2025 MRS Spring Meeting, Seattle, WA, USA / Apr. 2025**
Graduate Student Award Talk & Poster: “Advancing Quantum Material Discovery through Structural Constraint Integration in Generative Models”
2. **Competing Orders in Quantum Materials, New Haven, CT, USA / Mar 2024**
Oral: “Advancing Quantum Material Discovery through Structural Constraint Integration in Generative Models”
3. **2024 MRS Fall Meeting, Boston, MA, USA / Dec. 2024**
Oral: “Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates”

Oral: "Large Language Model-Guided Prediction Toward Quantum Materials Synthesis"

4. **Northeast Quantum Forum 2024**, *Durham, NH, USA / Oct 2024*

Invited Talk: "Machine-learning assisted study of quantum material properties and quantum material discovery"

5. **Quantum Materials for Emergent Applications in Quantum Science**, *Telluride, CO, USA / June 2024*

Invited Talk: "Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates"

6. **2024 ACS Spring Meeting** *New Orleans, LA, USA / Mar. 2024*

Poster: "Virtual Node Augmented Machine Learning for Material Property Prediction"

7. **2023 MRS Spring Meeting** *San Francisco, CA, USA / Apr. 2023*

Poster: "Virtual Node Augmented Machine Learning for Material Property Prediction"

8. **2022 MRS Fall Meeting** *Boston, MA, USA / Nov-Dec. 2022*

Poster: "Machine Learning Gamma Phonons from Atomic Coordinates"

9. **2022 MRS Spring Meeting** *(online) / May. 2022*

Oral: "Application of Radiation Detection Materials for Radiation Mapping with Machine Learning"

10. **22nd International Society of Magnetic Resonance Conference** *(online) / Aug. 2021*

Poster: "Theoretical explanation of new solvent suppression scheme with adiabatic pulse and application for solid-state NMR experiments"

11. **The 59th Annual Meeting of the NMR Society of Japan**, *Gunma, Japan / Nov. 2020*

Poster Presentation: "Development of Suppression Pulse with Adiabatic Pulse and Application for Biological Solid-State NMR"

12. **The 58th Annual Meeting of the NMR Society of Japan**, *Kanagawa, Japan / Nov. 2019*

Poster Presentation: "Development of Suppression Pulse with Adiabatic Pulse for Solid-State NMR Experiment"